

Antonio J. Torres

Personal Details

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Education

- 2019-2023 **PhD in Mathematics**, Universidad Nacional Autónoma de México (UNAM), Juriquilla, Mexico.
Advisor: Dr. Deborah Oliveros.
Thesis: Ramsey Type Theorems on Simplicial Complex Nerves.
- 2015-2018 **Master in Applied Mathematics**, Universidad Autónoma de San Luis Potosí (UASLP), Mexico.
Advisor: Dr. Gelasio Salazar Anaya.
Thesis: Blowing Circular Sequences to Search for Drawings with Few Crossings.
- 2008-2013 **Bachelor in Applied Mathematics**, Universidad Autónoma de Querétaro (UAQ), Mexico.
Advisor: Dr. Deborah Oliveros.
Thesis: A (p, q) -problem for Linear and Cyclical Orders.

Research

- 2023 De Loera, J. A., Jaramillo-Rodriguez, E., Oliveros, D., & **Torres, A. J.** A model for birdwatching and other chronological sampling activities. *The American Mathematical Monthly*, 130(6), 541–558. <https://doi.org/10.1080/00029890.2023.2188036>
- 2025 Oliveros, D., Roldán, E., Soberón, P., & **Torres, A. J.** Tverberg partition graphs. *SIAM Journal on Discrete Mathematics*, 39(2), 863–880. <https://doi.org/10.1137/23M1610938>
- 2026 De Loera, J. A., López-Campos, G., & **Torres, A. J.** On the number of vertices in hyperplane cross-sections of convex polytopes. Accepted for publication in *Discrete & Computational Geometry (DCG)*; to appear. Preprint available on arXiv: <https://arxiv.org/abs/2412.12419>.
- 2024 Oliveros, D., & **Torres, A. J.** From word-representable graphs to altered Tverberg-type theorems. *Discrete & Computational Geometry*. <https://doi.org/10.1007/s00454-025-00718-5>.
- 2026 Averkov, G., Brose, A. E., De Loera, J. A., López-Campos, G., & **Torres, A. J.** On Lattice Diameter Segments: Algorithms and Structure. In S. S. Dey, M. Di Summa, & D. Salvagnin (Eds.), *Integer Programming and Combinatorial Optimization (IPCO 2026)*, Lecture Notes in Computer Science, vol. 16588. Springer, Cham. https://doi.org/10.1007/978-3-032-28691-8_15

In Progress

- D. Oliveros, S. G. Gómez-Galicia, and A. J. Torres, *Crossing-Free Partition Graphs*. In preparation.

Professional Experience

Postdoctoral Positions

- 2025– **Postdoctoral Researcher**, Smith College, Northampton, MA, USA.
Advisor: Dr. Ileana Streinu.
Research in discrete and computational geometry; collaboration with the Geometry Group.
- 2023–2025 **Chancellor’s Postdoctoral Fellow**, University of California, Davis, USA.
Advisor: Dr. Jesús A. De Loera.
Research in combinatorics, discrete geometry, and applications to lattice polytopes.

Teaching Experience

- 2025– **Teaching Assistant**, Smith College, USA
Courses: Data Structures, Applied Algorithms, Discrete and Computational Geometry.
- 2024–2025 **Instructor**, University of California Davis, USA
Courses: Mathematics and Computers.
- 2022–2024 **Instructor**, Mexican Mathematics Olympiad, Querétaro Delegation.
Courses: Combinatorics, Geometry (high school level).
- 2018–2019 **Instructor**, Universidad Politécnica de Querétaro (UPQ), Mexico.
Courses: Calculus, Mathematical Functions, Linear Algebra.
- 2014 **Instructor**, Universidad Autónoma de Querétaro (UAQ), Mexico.
Role: Advisor for university admission tests.

Industry Experience

- 2017 - 2018 **Data Analyst**, Out Helping (Recruiting company), Mexico.
Role: Administration and management of the company’s database.

Visiting Positions

- Oct-Dec 2022 **Max Planck Institute**, Leipzig, Germany.
Collaborator: Dr. Erika Roldan Roa.

Professional Service and Community Outreach

- Reviewer for *SIAM Journal on Discrete Mathematics*.
 - Reviewer for *Discrete Applied Mathematics*.
 - Reviewer for articles submitted to the volume *Women in Mathematics Latin America*, to appear in the Springer-Birkhäuser *Trends in Mathematics* series.
- 2022–2024 Volunteer to prepare participants from the state of Querétaro for the Mexican Mathematics Olympiad.
- 2019–2023 Participant in various outreach events and fairs organized by the National Autonomous University of Mexico (UNAM).
- 2016–2023 Volunteer work to assist senior citizens in obtaining their high school diplomas.
- 2014 Volunteer teaching students of all levels in the community of Loma Bonita, Oaxaca, Mexico, during the teachers’ strike.

Awards

- 2023–2025 Chancellor’s Fellow Award.
- 2019–2023 National Scholarship for Graduate Studies, CONAHCYT.

2025 Winner of the academic poster presentation competition in the Latinx in the Mathematical Sciences Conference, Los Angeles, USA

Conferences and Workshops

Contributions to Event Organization

Sept 2023 **XII Latin-American Algorithms, Graphs and Optimization Symposium**, Huatulco, México.

Invited Research Conferences

Jul 2025 **Discrete, Convex, and Combinatorial Geometry Fest**, Calgary, Alberta, Canada

Nov 2024 **Canadian Mathematical Society Winter Meeting**, Kwantlen Polytechnic University, Richmond, British Columbia, Canada

Jun 2023 **New Trends from Classical Theorems in Geometry**, CMO Oaxaca and Banff International Research Station, Oaxaca, Mexico

Jun 2023 **Interplay Between Geometric Analysis and Discrete Geometry**, CMO Oaxaca and Banff International Research Station, Oaxaca, Mexico

Oct 2019 **Helly and Tverberg Type Theorems, workshop**, CMO Oaxaca and Banff International Research Station, Oaxaca, Mexico

Speaker

May 2025 **AMS Spring Western Sectional Meeting**, San Luis Obispo, USA

Mar 2025 **Latinx in the Mathematical Sciences Conference**, UCLA, Los Angeles, USA

Dec 2024 **Bay Area Discrete Math Conference**, UC Berkeley, Berkeley, USA

Jul 2024 **9th Postdoctoral Research Symposium**, UC Davis, Davis, USA

Apr 2023 **AMS Spring Central Sectional Meeting**, Ohio, USA

Jul 2022 **Latinx in the Mathematical Sciences Conference**, session chair, Los Angeles, USA

Oct 2019 **LII National Congress of the Mexican Mathematical Society**, Monterrey, Mexico

Mar 2018 **XXXIII Víctor Neumann-Lara Colloquium on Graph Theory**, Havana, Cuba

Oct 2013 **XLV National Congress of the Mexican Mathematical Society**, Mérida, Mexico

Oct 2012 **XLV National Congress of the Mexican Mathematical Society**, Mexico

Participant

Jun 2024 **Recent Progress on Optimal Point Distributions**, ICERM, Providence, USA

Apr 2022 **XXXVI Víctor Neumann-Lara Colloquium on Graph Theory**, online

Aug 2021 **Encuentro Nacional de Computación 2021**, online

Jun 2021 **Latinx Mathematicians Research Community (LMRC)**, online

Mar 2017 **XXXIII Víctor Neumann-Lara Colloquium on Graph Theory**, San Luis Potosí, Mexico

Oct 2016 **XLV National Congress of the Mexican Mathematical Society**, Aguascalientes, Mexico

Nov 2016 **II Winter School in Mathematics**, Oaxaca, Mexico

Jun 2016 **III Summer School in Combinatorics**, Juriquilla, Querétaro, Mexico

Oct 2015 **XLV National Congress of the Mexican Mathematical Society**, Hermosillo, Mexico

Mar 2015 **XXVIII Víctor Neumann-Lara Colloquium on Graph Theory**, Oaxaca, Mexico

Mar 2013 **XXVIII Víctor Neumann-Lara Colloquium on Graph Theory**, Morelia, Michoacán, Mexico

Mar 2012 **XXVII Víctor Neumann-Lara Colloquium on Graph Theory**, Tlaxcala, Mexico

Oct 2011 **XLIV National Congress of the Mexican Mathematical Society**, San Luis Potosí, Mexico

Mar 2011 **XXVI Víctor Neumann-Lara Colloquium on Graph Theory**, Pachuca, Mexico

Oct 2009 **XLII National Congress of the Mexican Mathematical Society**, Zacatecas, Mexico

—— Technical Skills

Programming: Python, R, Mathematica, Java

Math editors: LaTeX

O.S: Windows, Linux, macOS

AI tools: ChatGPT, Claude, DeepSeek, Copilot, Gemini (primarily for coding, project structuring, and mathematical visualization)

—— Projects & Links

GitHub: github.com/AntonioJTH — repositories on algorithms, computational geometry, and visualization tools

Math VisApps: Math visualization project — interactive applications for geometric and mathematical exploration

—— Languages

- Spanish
- English